

Q1. Concept of Software Quality and SQA. Importance in development

Software Quality

Software quality refers to the degree to which software:

- Meets **functional and non-functional requirements**
 - Is **reliable, efficient, maintainable, and user-friendly**
 - Is **free from defects**
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Software Quality Assurance (SQA)

SQA is a **systematic approach** to ensure quality throughout the **software development life cycle (SDLC)**.

- Focuses on **process improvement**
 - Prevents defects rather than just detecting them
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Importance of SQA

1. **Reduces defects early**
 2. **Improves reliability and performance**
 3. **Enhances customer satisfaction**
 4. **Reduces maintenance cost**
 5. **Ensures standardization**
 6. **Improves team productivity**
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Q2. Elements contributing to SQA

1. **Standards and Procedures**
 - Coding standards, documentation rules
2. **Reviews and Audits**
 - Peer reviews, technical inspections
3. **Testing**
 - Unit, integration, system testing
4. **Configuration Management**
 - Version control and change tracking
5. **Defect Tracking**
 - Logging and resolving bugs
6. **Training**

- Skill development for quality practices
 - 7. **Process Improvement**
 - Continuous refinement using feedback
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Q3. SQA tasks during SDLC

Planning Phase

- Prepare SQA plan
- Define standards

Development Phase

- Conduct reviews (design/code)
- Ensure compliance with standards

Testing Phase

- Verify testing processes
- Monitor defect tracking

Maintenance Phase

- Analyze defects
 - Improve processes
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Q4. Role of SQA in improving reliability & performance

Reliability

- Ensures fewer defects
- Improves system stability
- Uses testing and reviews to prevent failures

Performance

- Ensures efficient use of resources
- Identifies bottlenecks through testing
- Optimizes system behavior

☞ Example: Load testing ensures software performs well under heavy usage.

Q5. Static vs Dynamic Quality Assurance

Aspect	Static QA	Dynamic QA
Definition	Without executing code	With execution
Techniques	Code review, inspections	Testing
Goal	Prevent defects	Detect defects
Stage	Early (design/coding)	After development

Example

- Static: Reviewing code manually
- Dynamic: Running test cases

Q6. Statistical Software Quality Assurance (SSQA)

SSQA uses **statistical techniques** to measure and improve quality.

Steps

1. Collect defect data
2. Classify defects
3. Analyze frequency
4. Identify high-risk areas
5. Improve processes

Benefits

- Data-driven decisions
- Identifies critical modules
- Reduces defect rates

☞ Example: If most bugs occur in one module → improve that module.

Q7. Six Sigma in Software Engineering

Six Sigma aims to reduce defects to **3.4 defects per million**.

DMAIC Approach

1. Define problem
 2. Measure performance
 3. Analyze causes
 4. Improve process
 5. Control future performance
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Contribution to Quality

- Reduces variability
 - Improves consistency
 - Enhances customer satisfaction
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Q8. ISO 9000 Quality Standards

ISO 9000 is a set of **international standards** for quality management.

Key Features

- Focus on process quality
 - Documentation and control
 - Continuous improvement
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Role in Software Quality

- Ensures consistency
 - Improves credibility
 - Enhances customer trust
 - Provides structured processes
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Q9. Ishikawa's 7 Basic Tools

1. Cause-and-Effect Diagram (Fishbone)

- Identifies root causes of problems

2. Check Sheets

- Structured data collection

3. Control Charts

- Monitor process variation

4. Histograms

- Show data distribution

5. Pareto Chart

- Focus on major issues (80/20 rule)

6. Scatter Diagram

- Shows relationship between variables

7. Flowcharts

- Visualize process flow

Q10. Pareto Diagram and its role

Definition

A Pareto diagram is a **bar chart** showing problems in decreasing order of importance.

Principle

- Based on **80/20 rule**
- 80% problems come from 20% causes

Role

- Identifies critical issues
- Helps prioritize problem-solving
- Improves quality efficiently

☞ Example: Most bugs come from few modules → focus on them.

Q11. Role of Checklists in SQA

Definition

A checklist is a list of items used to ensure all steps are followed.

Uses

- Code reviews
 - Testing procedures
 - Audits
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Benefits

- Reduces human error
 - Ensures consistency
 - Improves process reliability
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Q12. Compare Six Sigma and Statistical SQA

Aspect	Six Sigma	Statistical SQA
Focus	Process improvement	Quality measurement
Approach	DMAIC methodology	Statistical analysis
Goal	Reduce defects to near zero	Identify defect trends
Nature	Structured methodology	Analytical technique
